Data Science for Public Policy

Clean Policy, Green Investment, and Country Context: A Cluster-Based Analysis

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Agenda

1. Goal of the study
2. Climate Policy Database
3. World Bank PPI Database
4. Regression analysis
5. Conclusion

- Appendix

Goal of the study

Research question

- How does the number of clean energy policies relate to green investment across countries?
- Does this relationship differ across country clusters defined by income and institutional quality?

Datasets

- Climate Policy Database
- World Bank Public Participation in Infrastructure (PPI)

Climate Policy Database

The Climate Policy Database

is a platform maintained by the NewClimate institute.

It tracks the implementation status of climate policies and identifies coverage gaps.

Policies with a clear climate change mitigation objective are included.

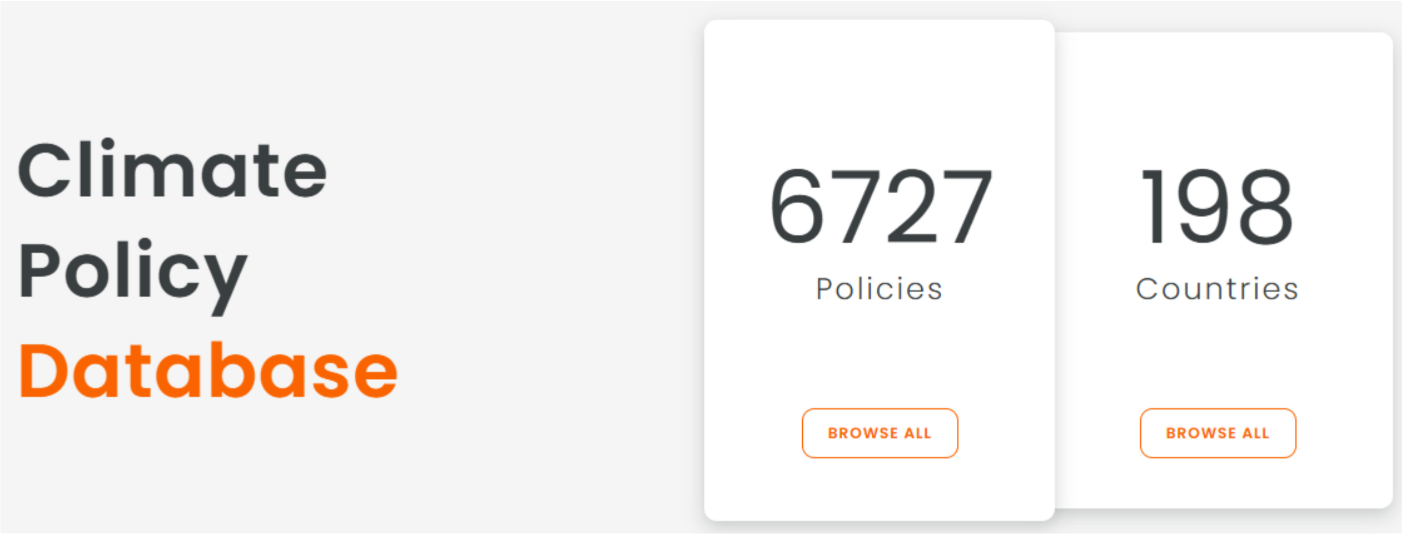
Coverage

42 countries including the EU are closely tracked, while the coverage for other countries is not exhaustive.

A policy can be a law, strategic document, a target, or any other policy document.

Availability

The data is openly available on the website and through their [Python API]

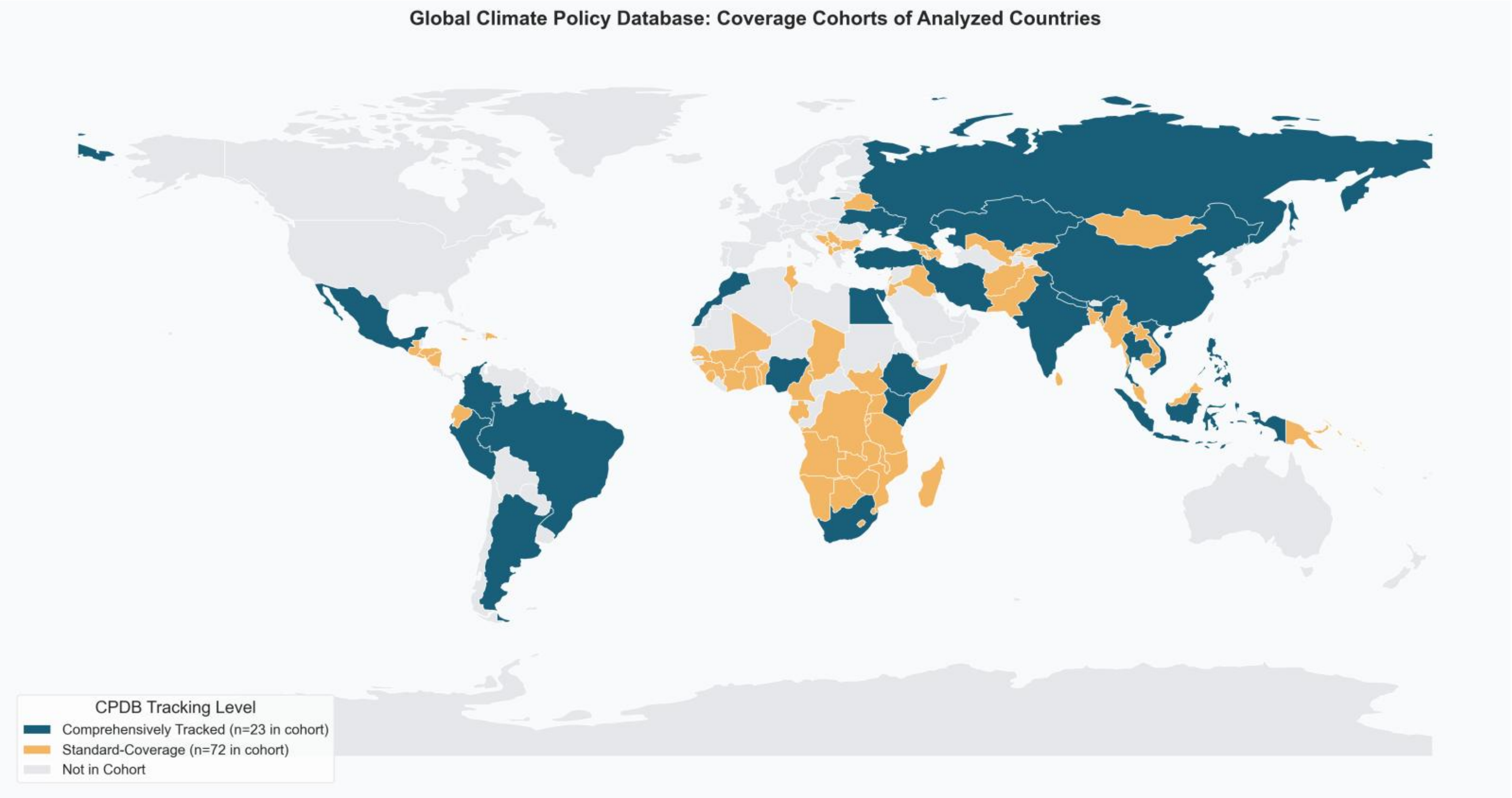


Our dataset from the Climate Policy Database

Using the API, we extracted all the relevant policies to our work.
By only considering the countries covered by the PPI data, we obtain a dataset of 95 countries

Coverage

Out of the 95 countries, 23 are in the “closely tracked” group, and have an average policy count of 88. The other 72 countries have an average of 8 policies. We acknowledge this disparity and show differentiated plots.

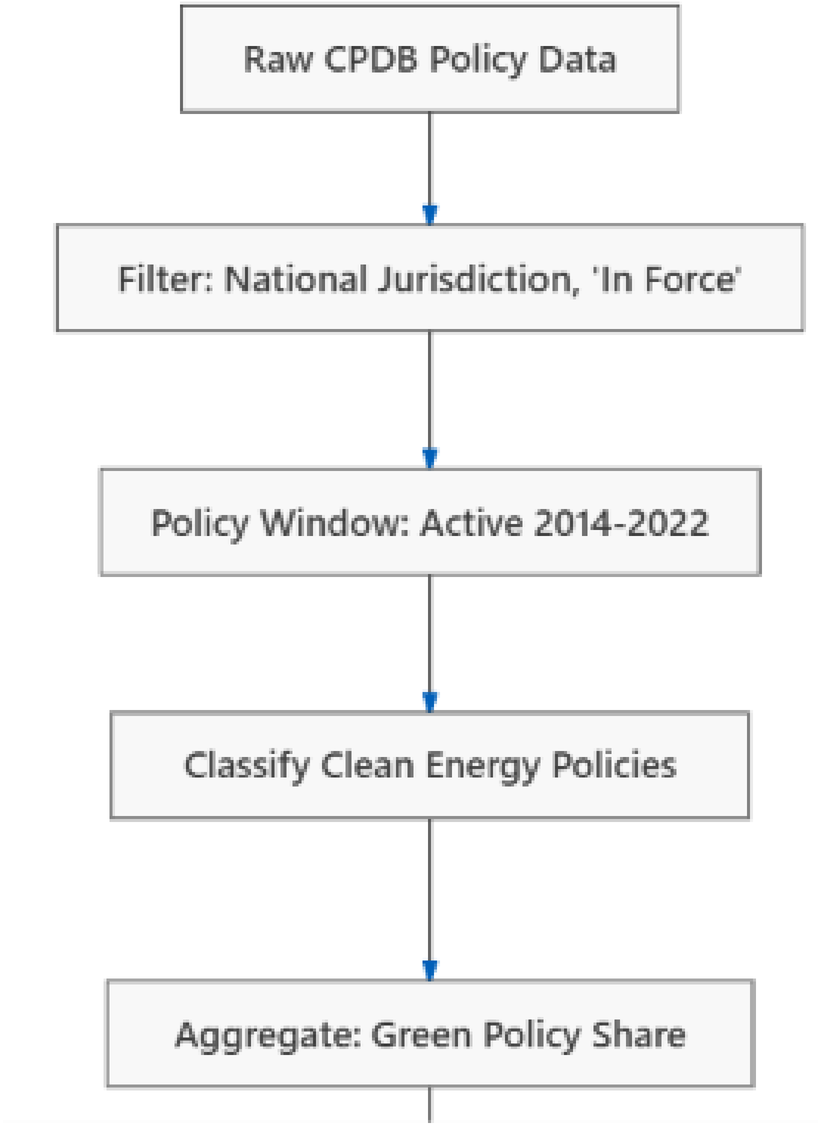


Our dataset from the Climate Policy Database

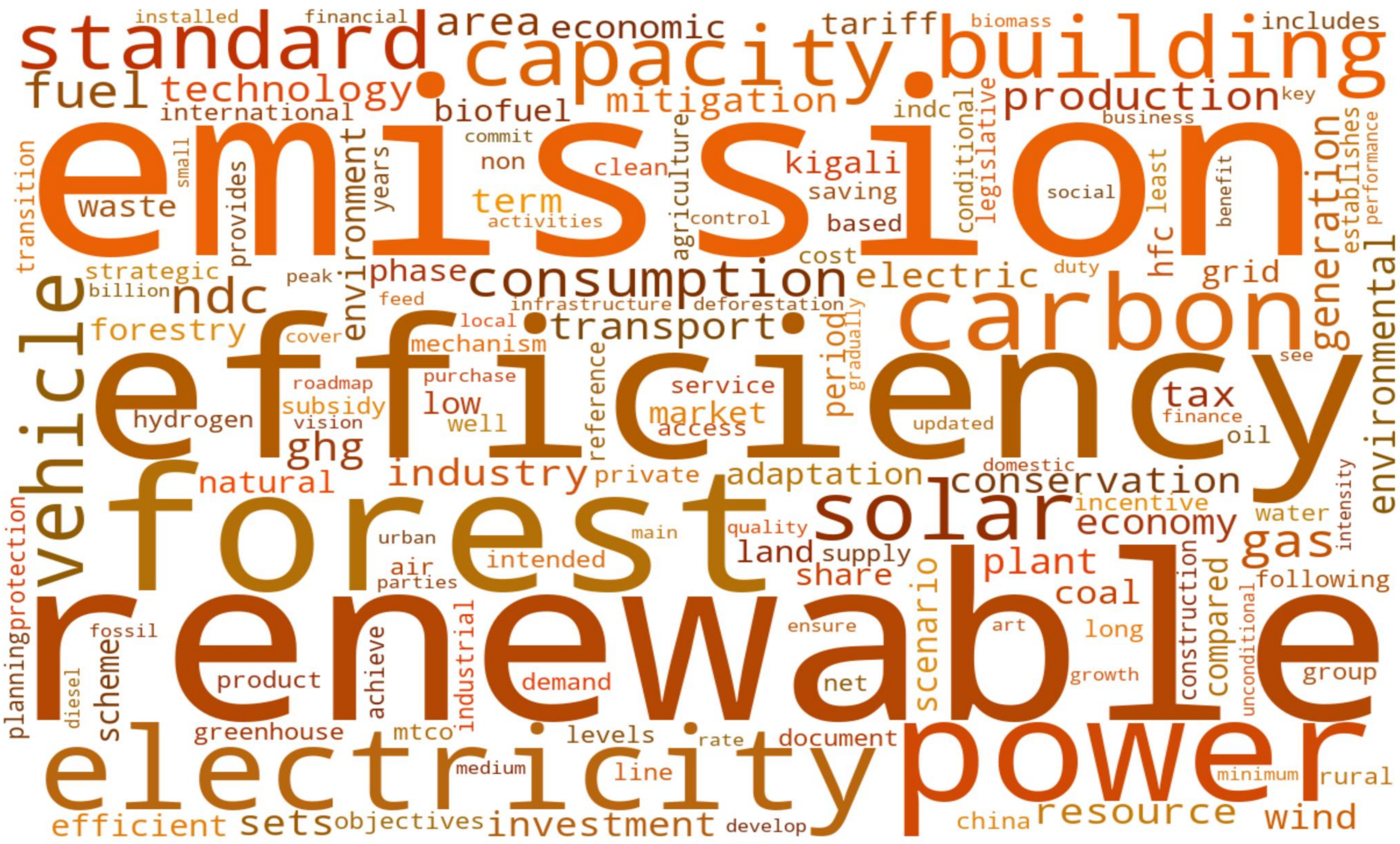
Filtering

We only want to consider policies related to clean energy. After exploring the dataset and consulting different articles related to clean energy taxonomy, we extract clean policies.

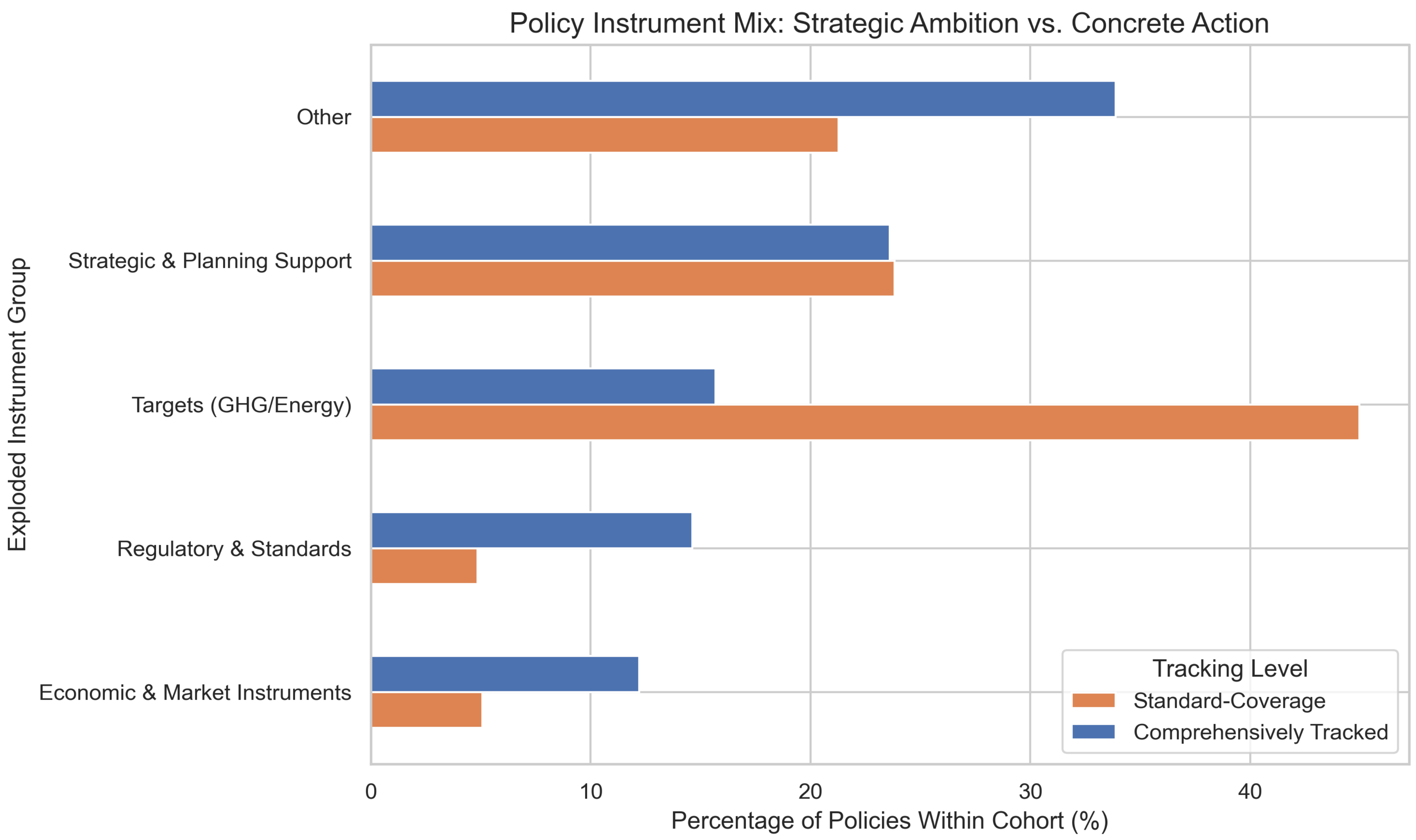
Specifically, the main filter is if policy type or sector contained terms like Renewables, Electricity, or Energy Efficiency



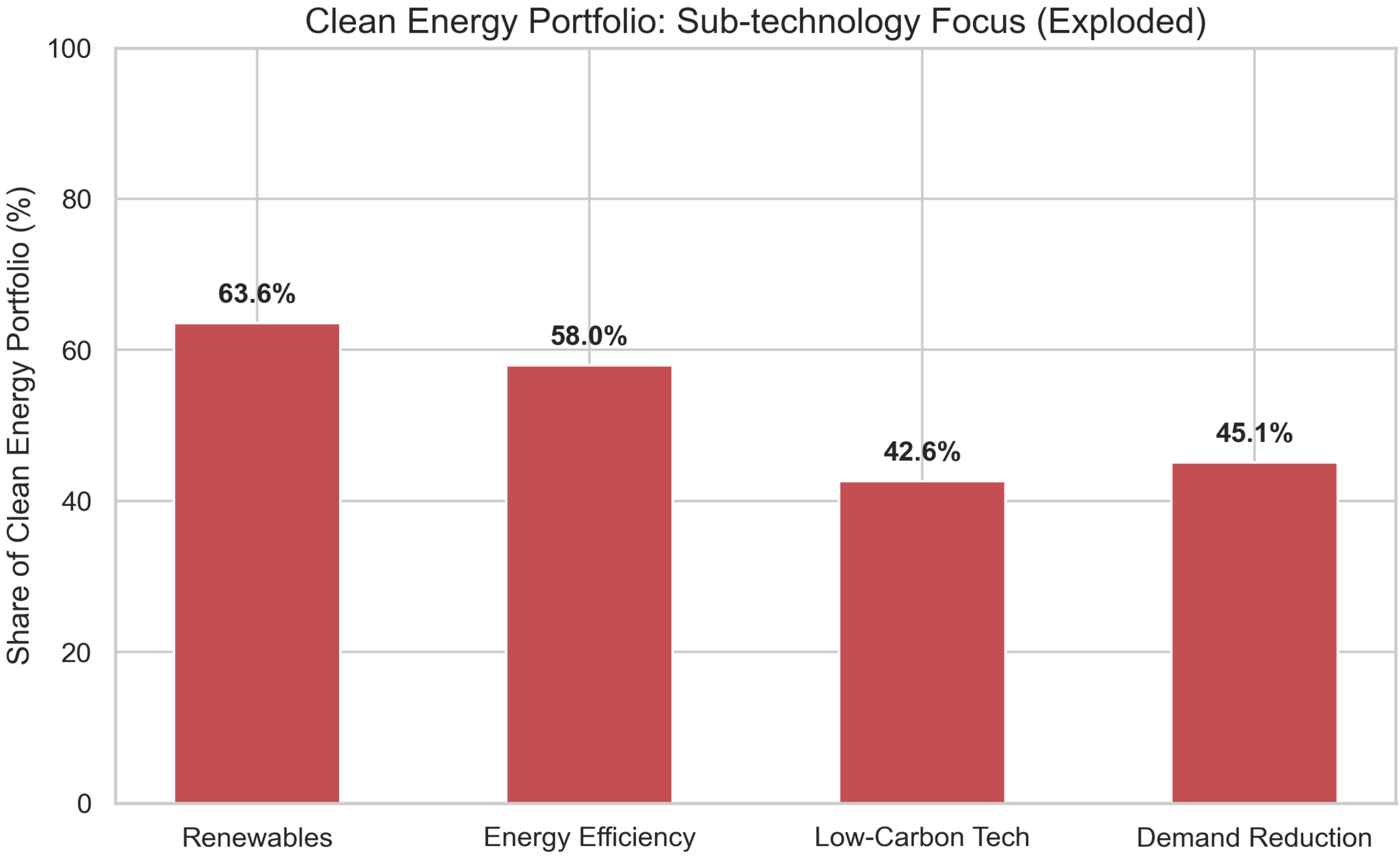
Text analysis: word cloud



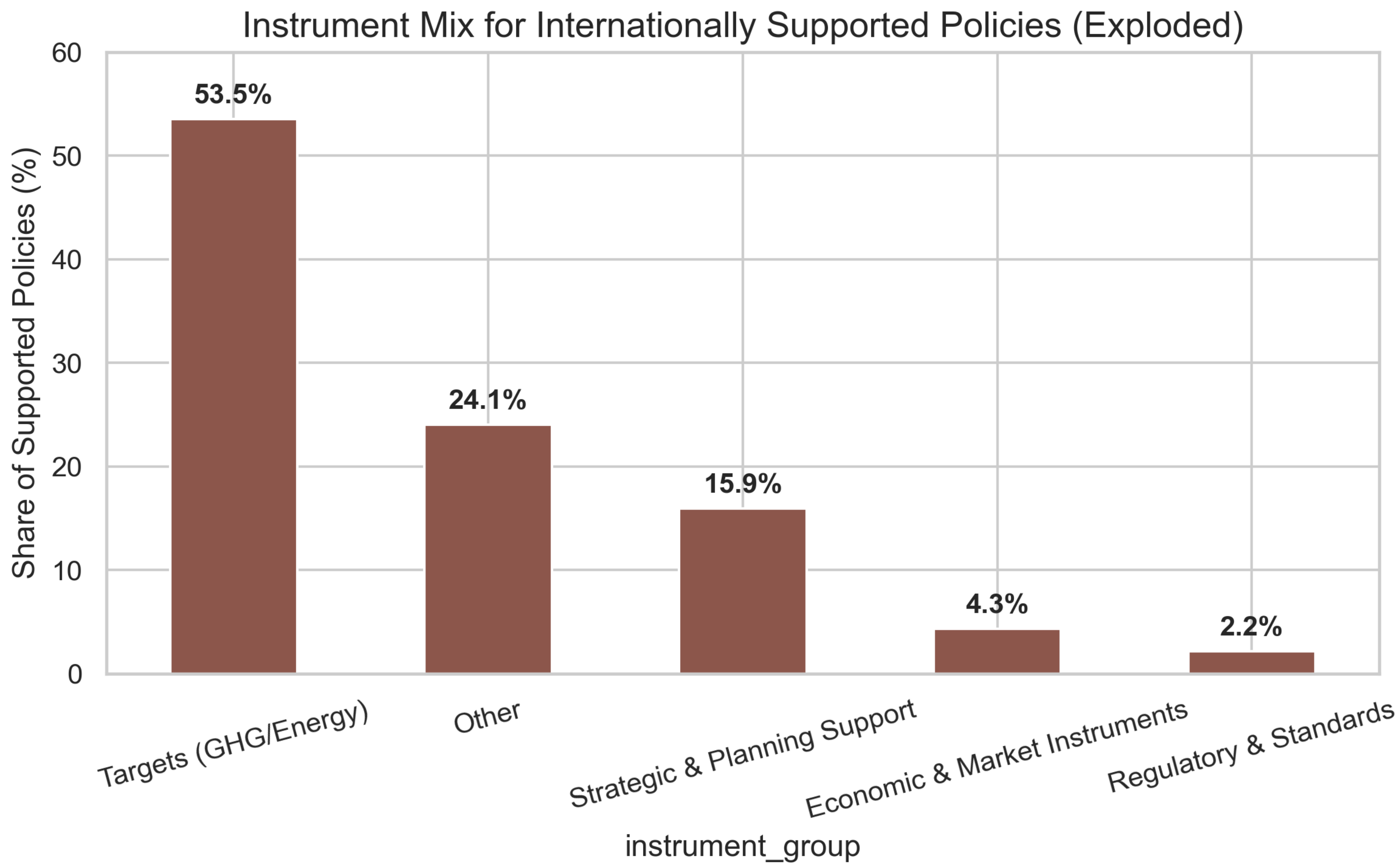
Policy instruments



Focus on clean energy policies: Technology type



Focus on internationally supported policies



World Bank PPI Database

World Bank PPI Database

- The World Bank PPI Database tracks private sector participation in infrastructure projects in **130 low and middle income countries from 2000-2024**
- Covers energy, transport, water, ICT, and municipal solid waste sectors
- Each project recorded at financial close with investment amount, technology type, sponsor origin, and sector

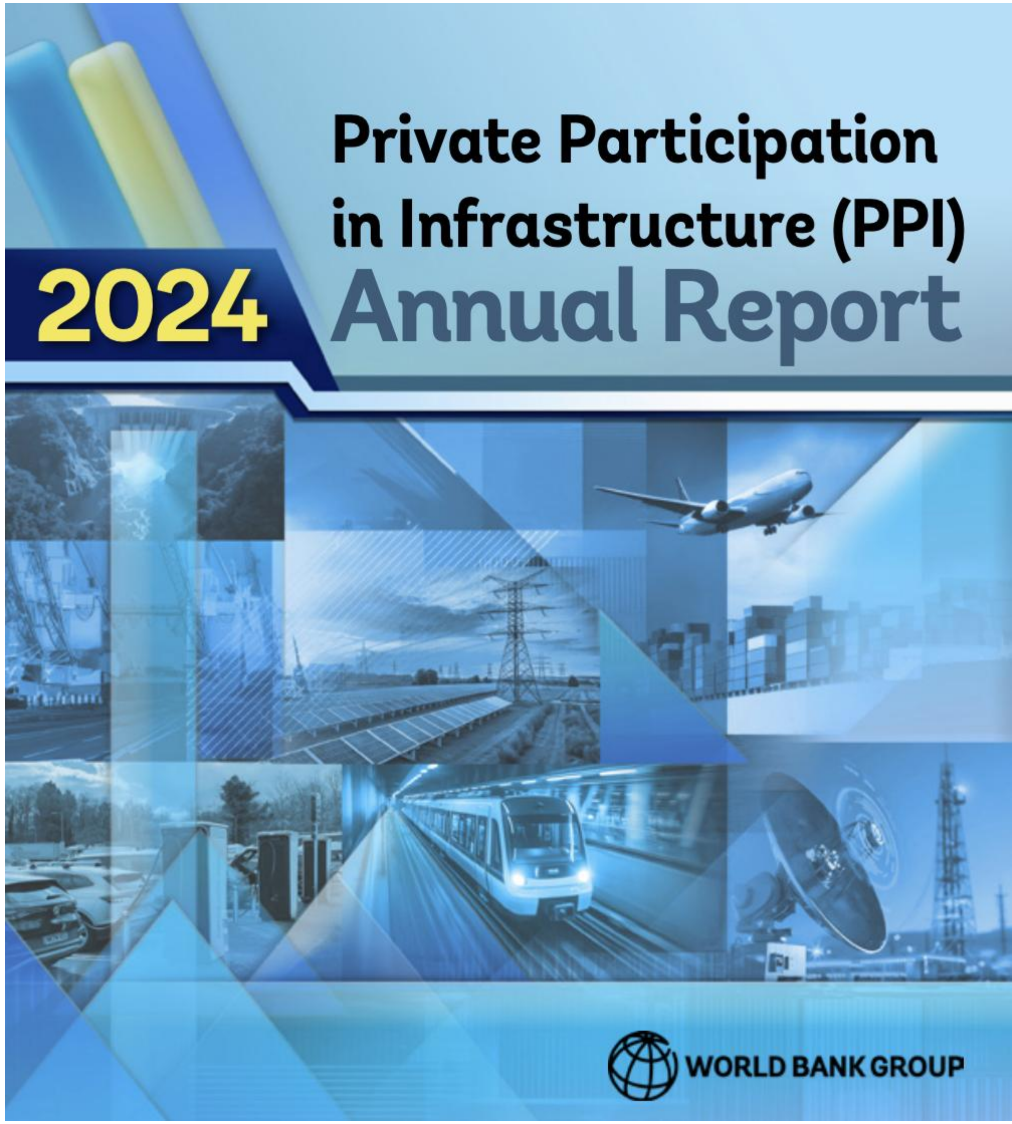


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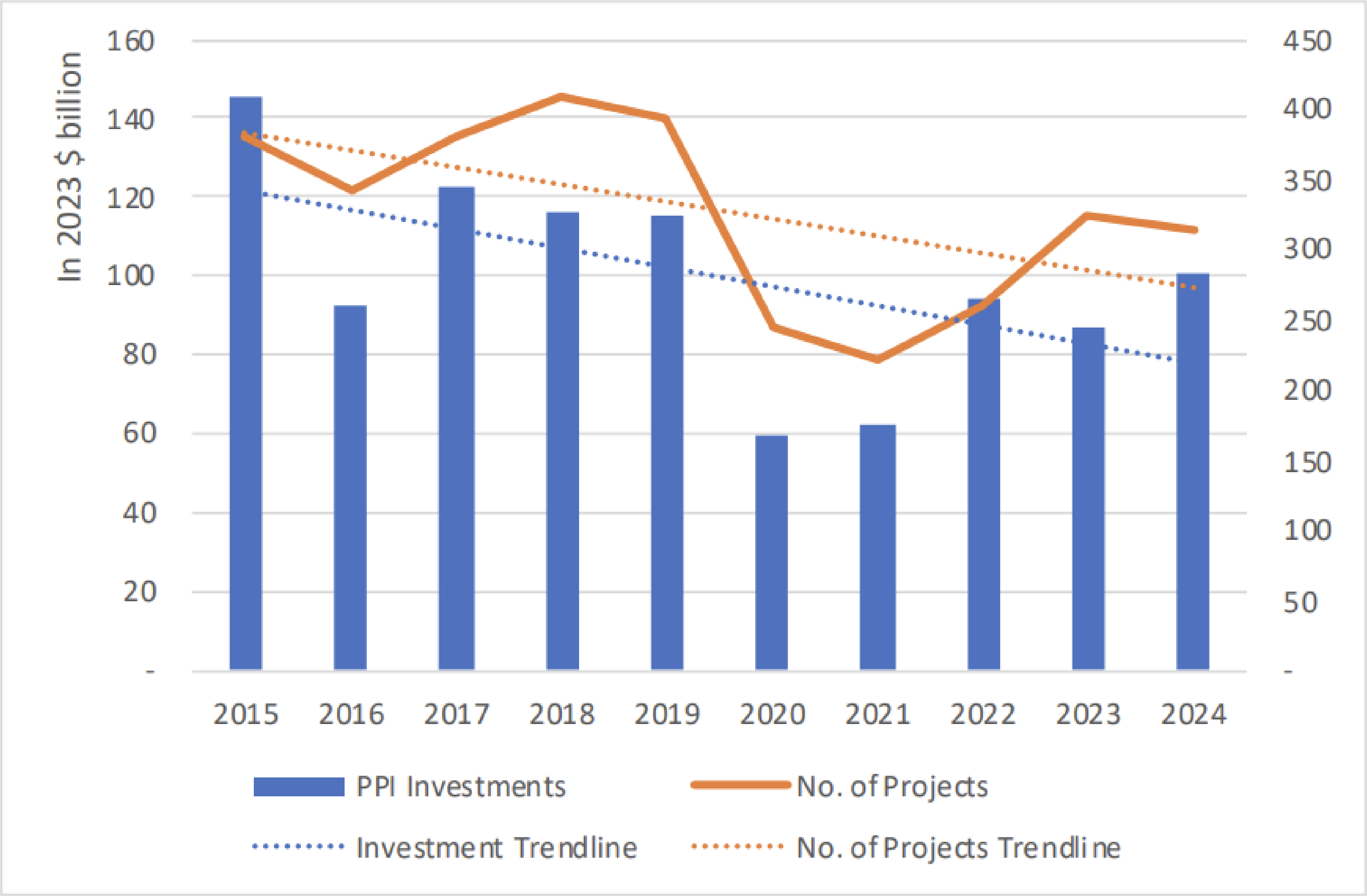
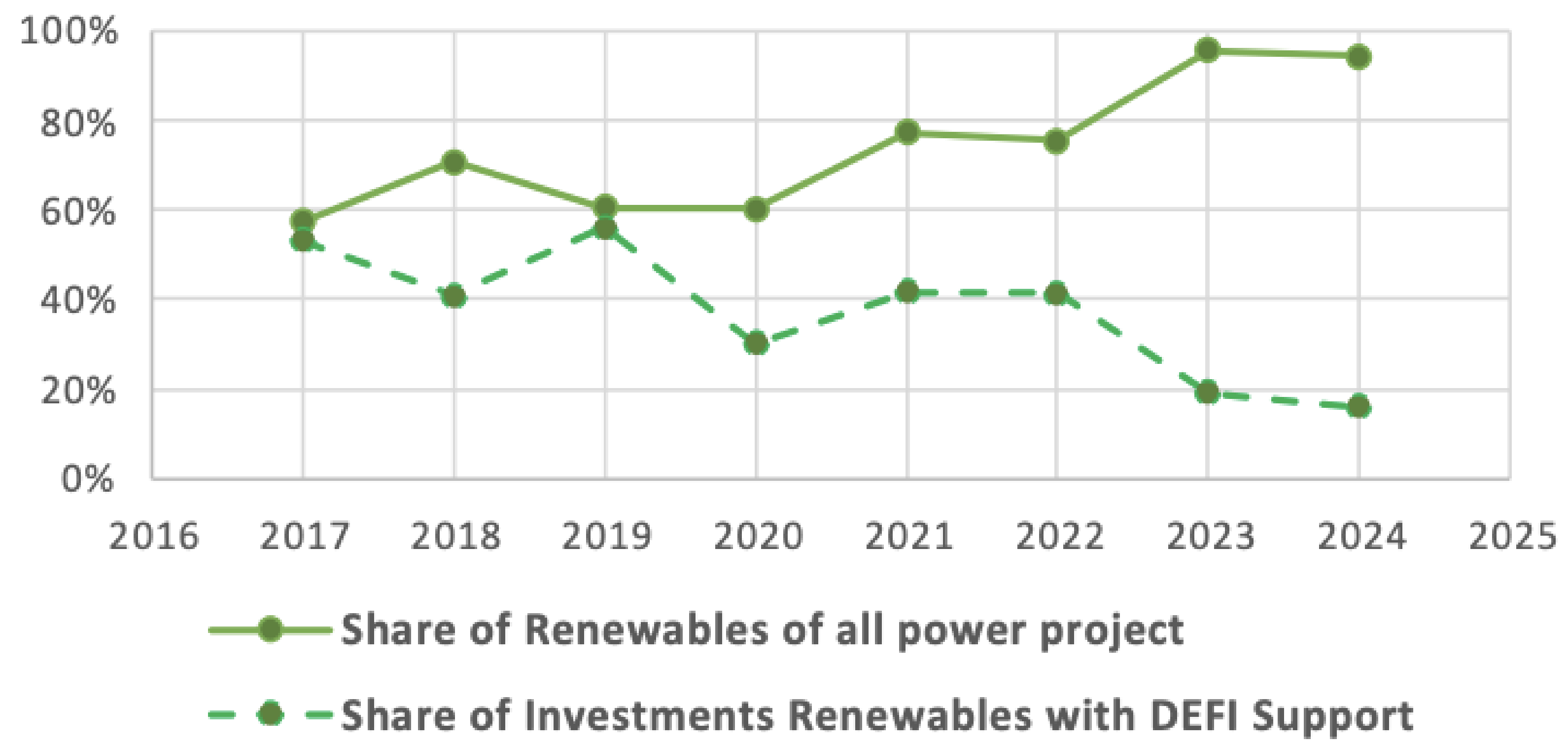
Figure 1: Investment Commitments in Infrastructure Projects with Private Participation in Low- and Mid-Income Countries, 2014– 2023

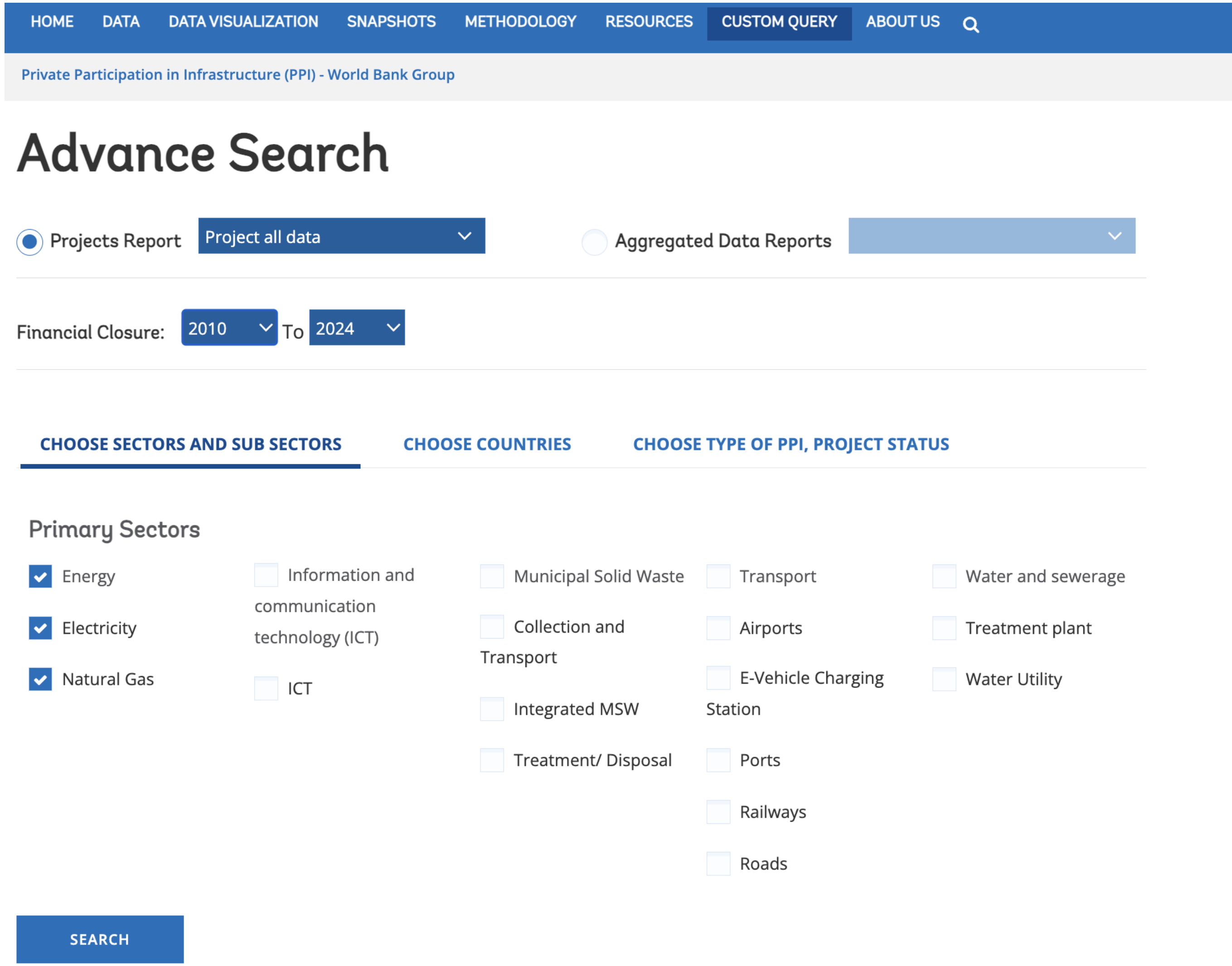


Renewable Energy Projects Dominate PPI Investments in Energy Proejcts



Custom Query on PPI Database Website

- Using the World Bank PPI Custom Query tool, extracted all energy sector projects with financial closure between 2010 and 2024



Our dataset from the PPI Database

- Filtered to the Electricity subsector only, removing Natural Gas transmission and distribution projects which cannot be classified as green or brown

- Focused on the Technology column, one of 45 available fields, to classify each project's generation source

Filter to Electricity Subsector

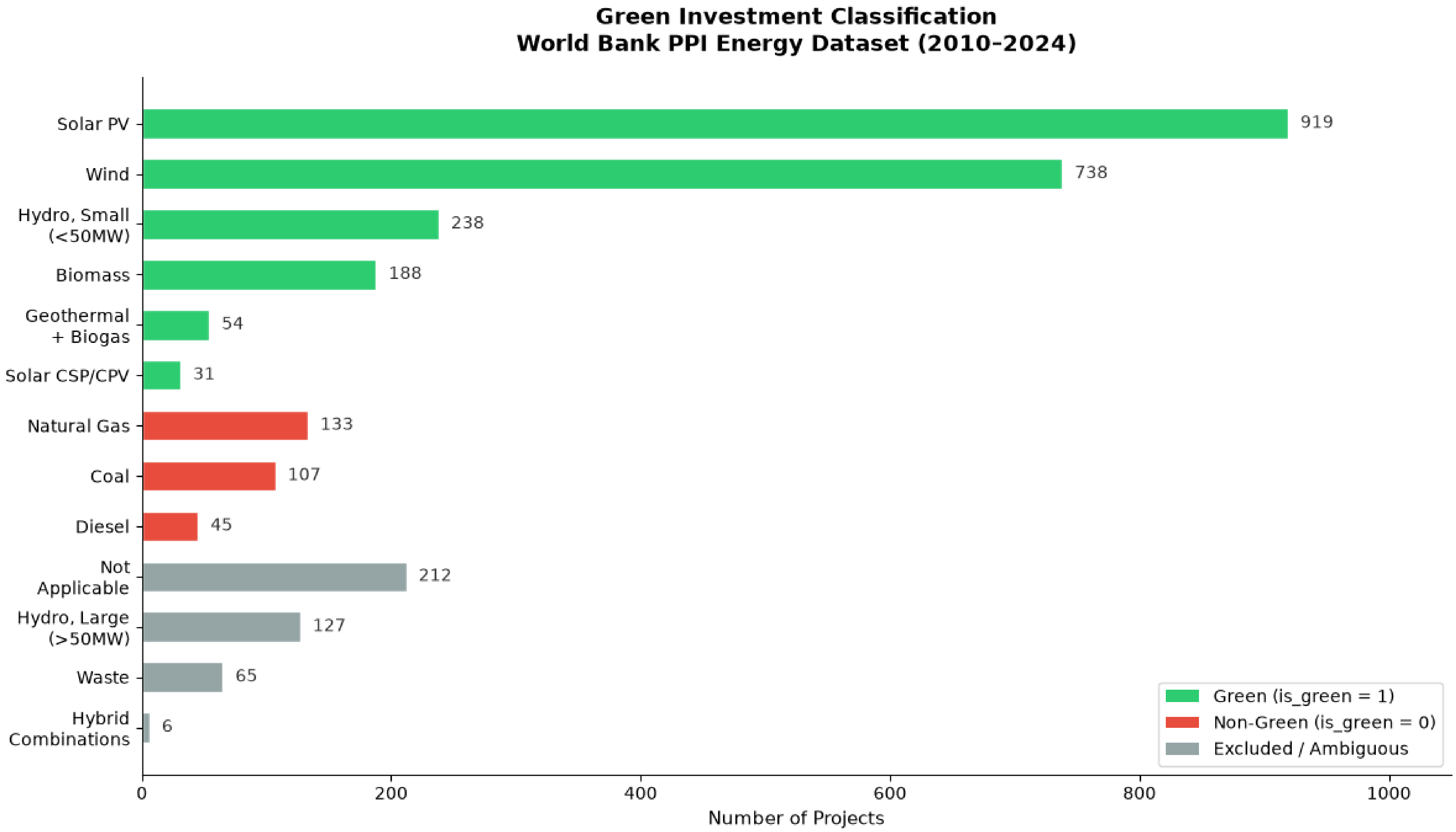
The raw data covers all energy PPI projects (electricity, pipelines, natural gas distribution, etc.). We keep only rows where `Subsector == 'Electricity'` since the analysis focuses on electricity generation and transmission investment.

```python
print('Shape before filter:', df.shape)

df_elec = df[df['Subsector'] == 'Electricity'].copy()

print('Shape after filter: ', df_elec.shape)
print('\nTechnology value_counts:')
print(df_elec['Technology'].value_counts())
```

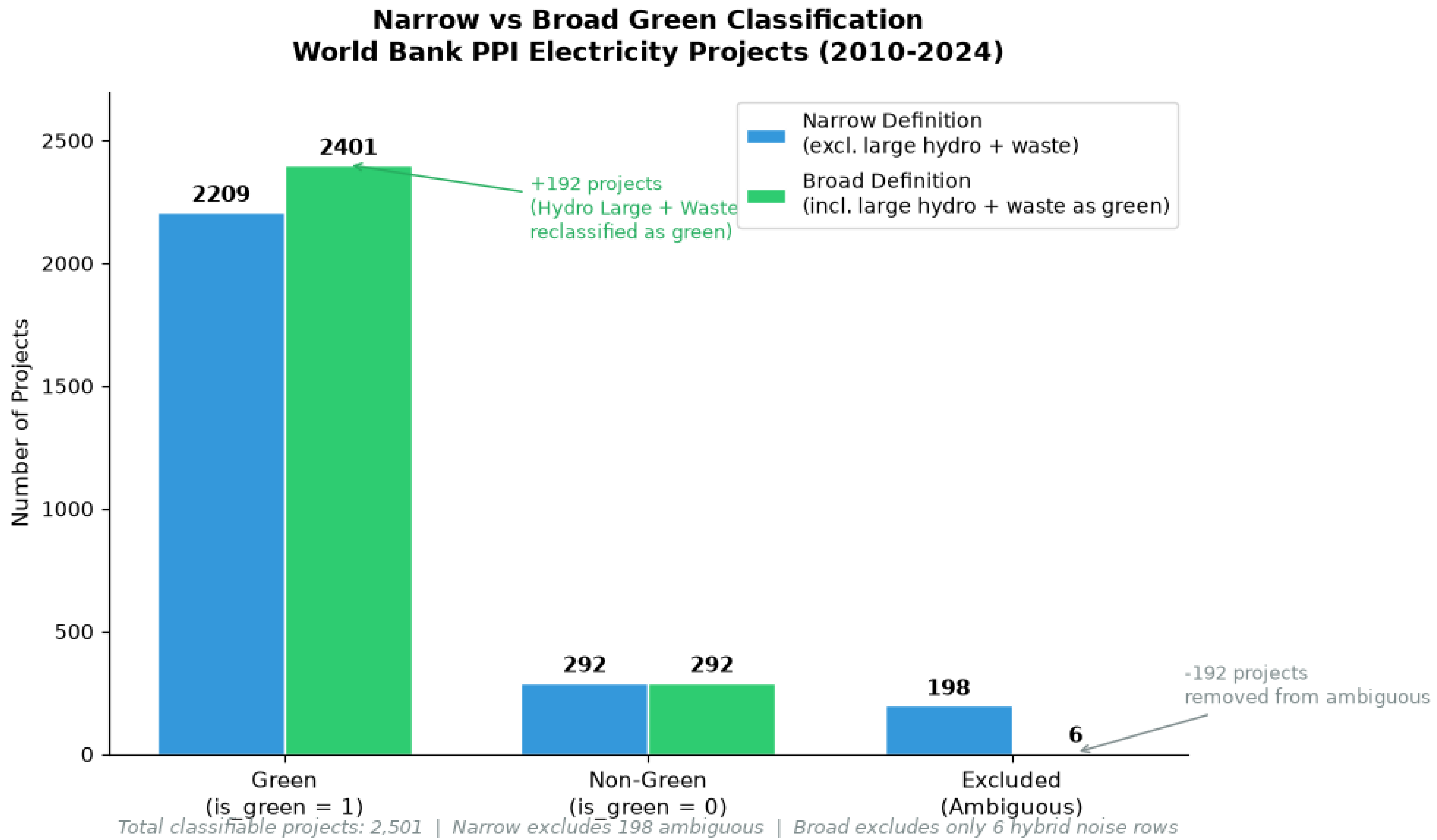
[2] Python

```
...
Shape before filter: (3008, 45)
Shape after filter:  (2951, 45)
```

Green Investment Classification



Narrow vs Broad Definition



Regression Analysis

Regression Analysis Overview

Research question

- How does the number of clean energy policies relate to green investment across countries?

Hypothesis 1: Higher number of clean energy policy can be associated with bigger amount of green investment

Number of Clean Energy Policy → Green Investment

- Does this relationship differ across country clusters defined by income and institutional quality?

Hypothesis 2: Countries with high GDP per capita and high policy implementation capability has more significant association

Methods

1. Clustering of countries based on

GDP per capita & **the Rule of Law index**

2. Regressions with the baseline model

$$\log(Green) = \beta_0 + \beta_1 Policy + \beta_2 GDP + \beta_3 Rule + \gamma Cluster + \varepsilon$$

3. Regressions with the interaction model

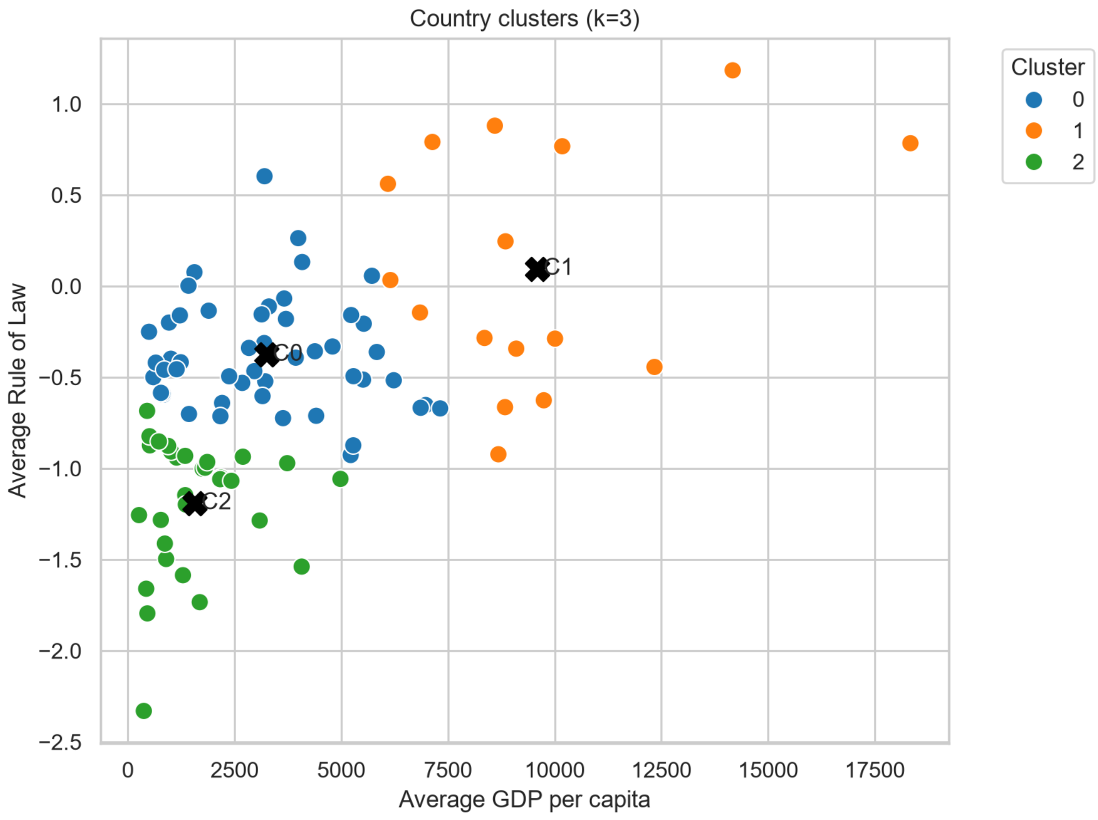
$$\log(Green) = \beta_0 + \beta_1 Policy + \beta_2 GDP + \beta_3 Rule + \gamma Cluster + \theta(Cluster \times Policy) + \varepsilon$$

Clustering

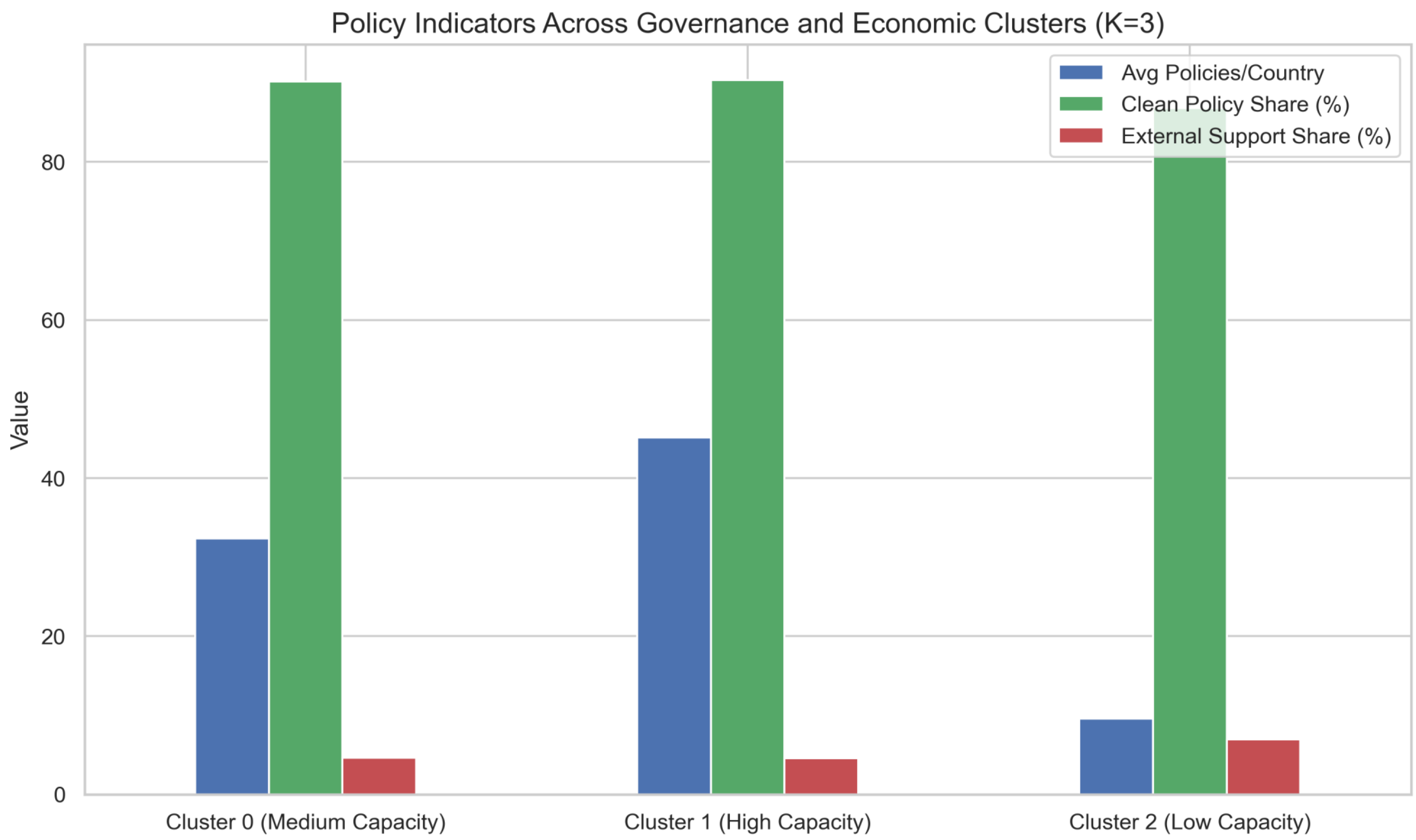
- Countries were grouped into three clusters using K-means on the standardized two variables (GDP per Capita & Rule of Law)

Cluster	N	Avg GDP per Capita	Avg Rule of Law	Avg Policy Count	Avg log (green investment)
0	49	3,242	-0.376	17.3	5.8
1	16	9,588	0.096	22.6	5.8
2	30	1,570	-1.190	5.5	5.0

Figure 1: cluster map



Cluster analysis



Regression Results

Regressions with the baseline model

$$\log(Green) = \beta_0 + \beta_1 Policy + \beta_2 GDP + \beta_3 Rule + \gamma Cluster + \varepsilon$$

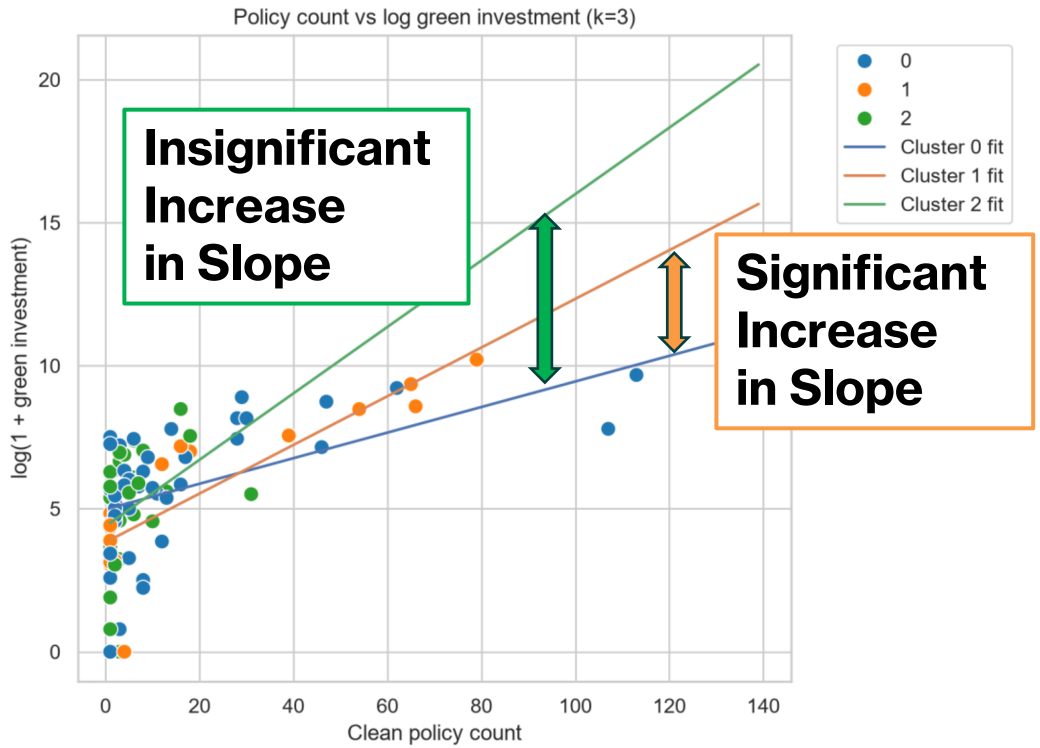
Term	Coefficient	Significance
Policy Count	$\beta 1 = 0.0545$	*** $(p<0.001)$

Overall positive significant correlation
Hypothesis 1 was verified

Regressions with the interaction model

$$\log(Green) = \beta_0 + \beta_1 Policy + \beta_2 GDP + \beta_3 Rule + \gamma Cluster + \theta(Cluster \times Policy) + \varepsilon$$

Figure 2:
policy count vs green investment with the interaction model



Term	Coefficient	Significance	Interpretation
Policy Count	$\beta 1 = 0.0449$	*** $(p<0.001)$	Reference coefficient (Cluster 0)
Cluster 1 * Policy Count	$\theta 1 = 0.0427$	** $(p=0.027)$	Statistically significant increase in C1 compared to C0
Cluster 2 * Policy Count	$\theta 2 = 0.0694$	$(p=0.568)$	No statistically significant difference between C0 and C2

Countries with high GDP per Capita and Rule of Law have high correlation
Hypothesis 2 was verified

Conclusion

Conclusion

- There is a **positive association** between the number of clean policies and green investment across countries.
- This relationship **varies by country context**. The interaction analysis suggests that the policy–investment relationship is strongest in countries with **higher income and stronger rule of law** (Cluster 1), while the difference is weaker for low-income countries.
- These findings imply that **institutional quality and economic development may enhance the effectiveness of climate policies** in attracting green investment.
- Because the analysis is based on **cross-sectional observational data**, the results should be interpreted as **associations rather than causal effects**.
- Future work will examine additional institutional and economic factors and apply more rigorous causal identification strategies.

References

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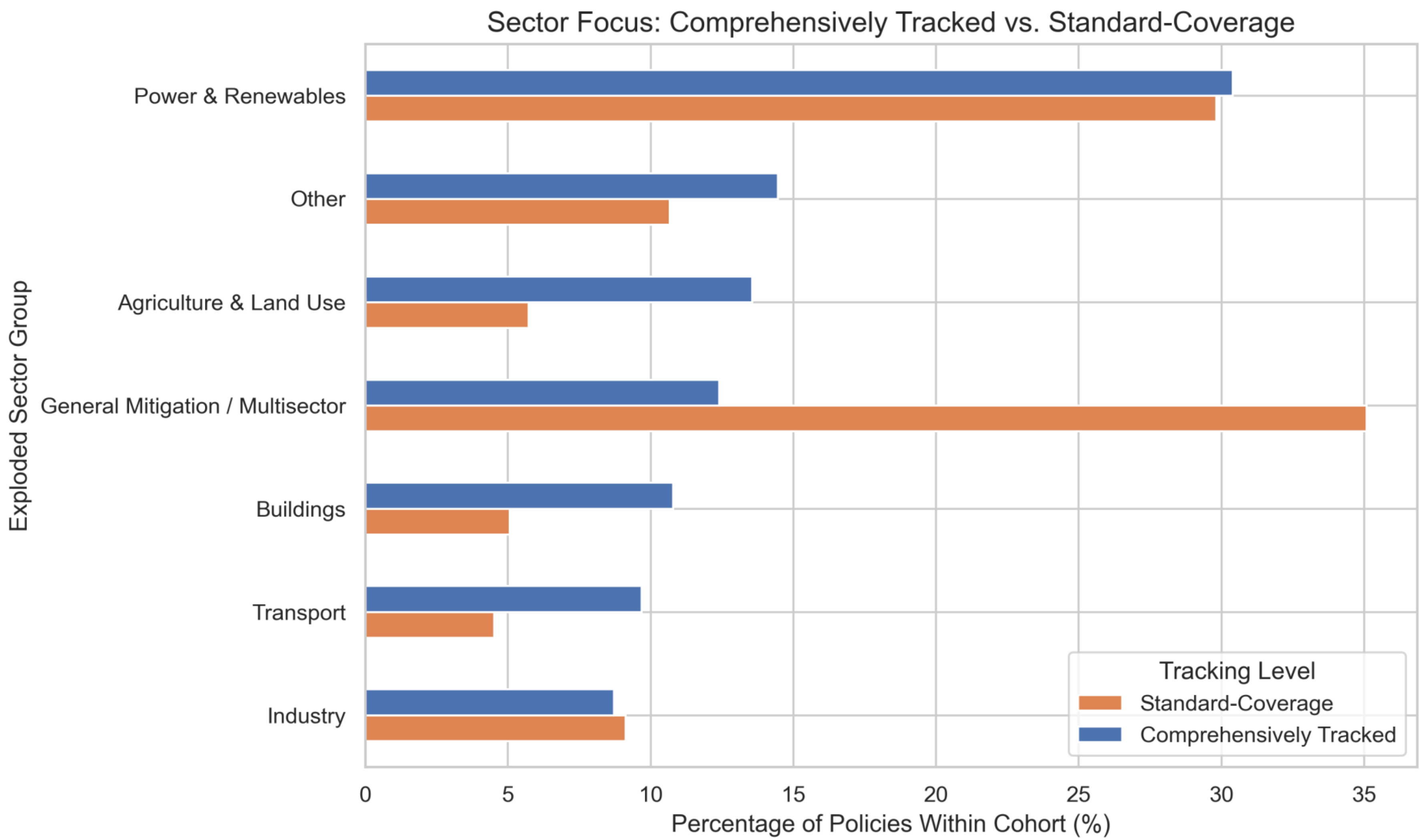
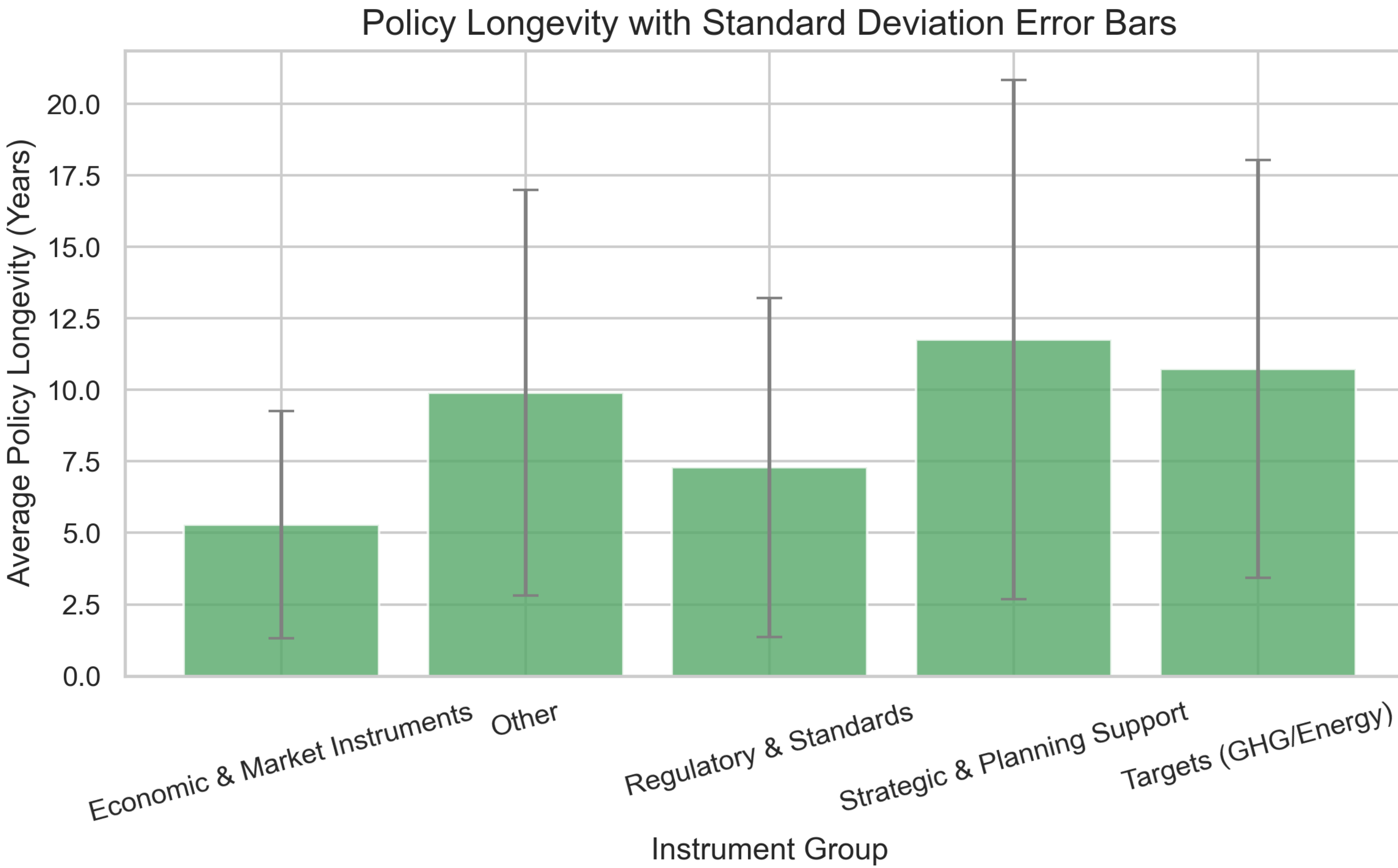
Appendix

Appendix: Policy sectors



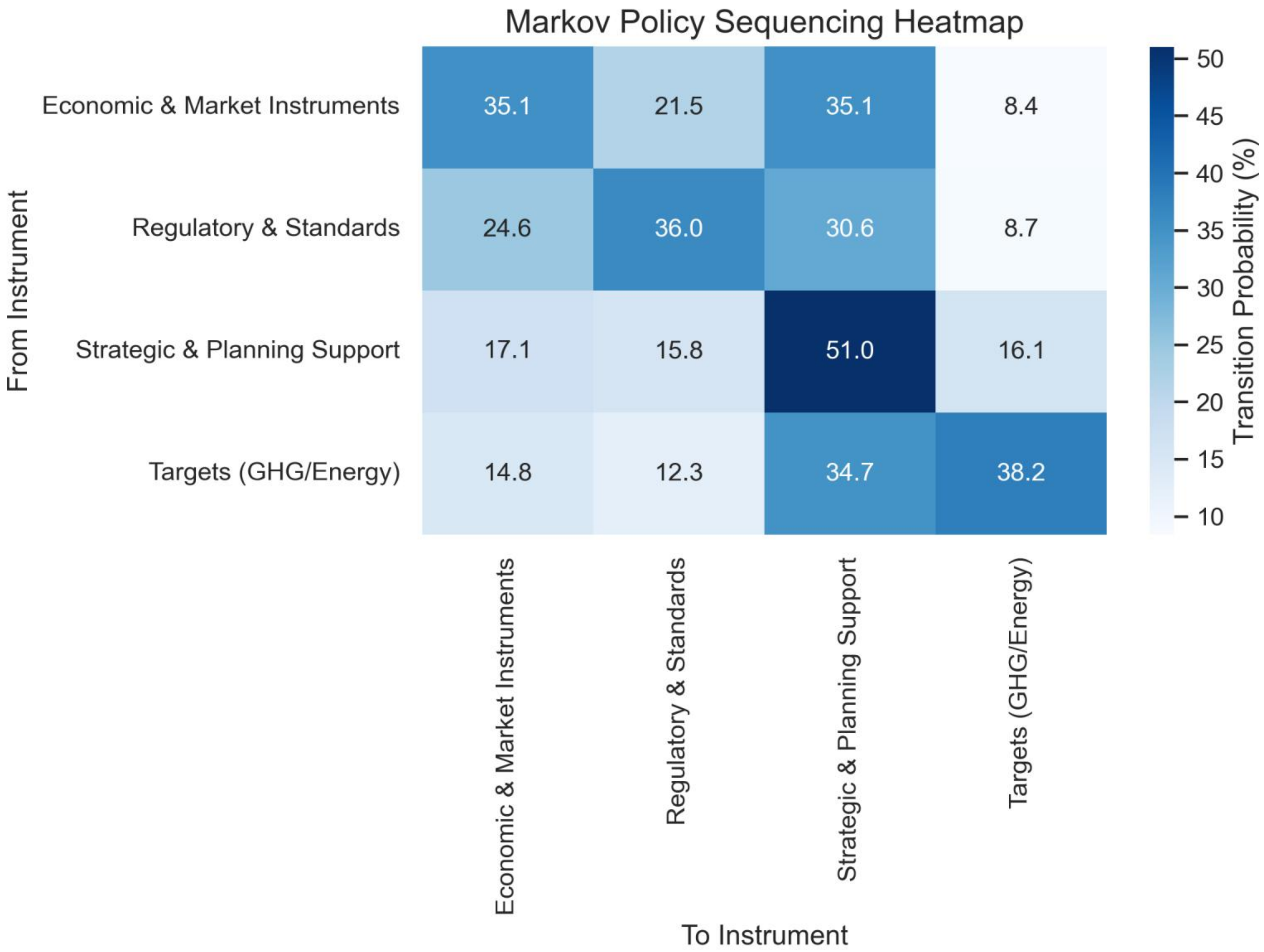
- Power & Renewables dominates policy portfolios for both groups (~37% for comprehensively tracked, ~29% for standard-coverage), while Buildings and Transport receive much less attention.
- Standard-coverage countries show a high concentration in "General Mitigation" (48.5%), indicating that their sparse records consist primarily of national "umbrella" strategies rather than sector-specific implementing regulations.

Appendix: Policy longevity



Strategic plans and targets have the longest lifespans (averaging 11.6 and 9.1 years, respectively), while regulatory standards and economic instruments are updated or retired much faster (averaging 5.4 and 5.5 years). Longevity can only be calculated for the ~17% of policies that populate both `start_date` and `end_date`. This highly incomplete sample may suffer from selection bias, as short-term or temporary policies are more likely to have explicit end dates.

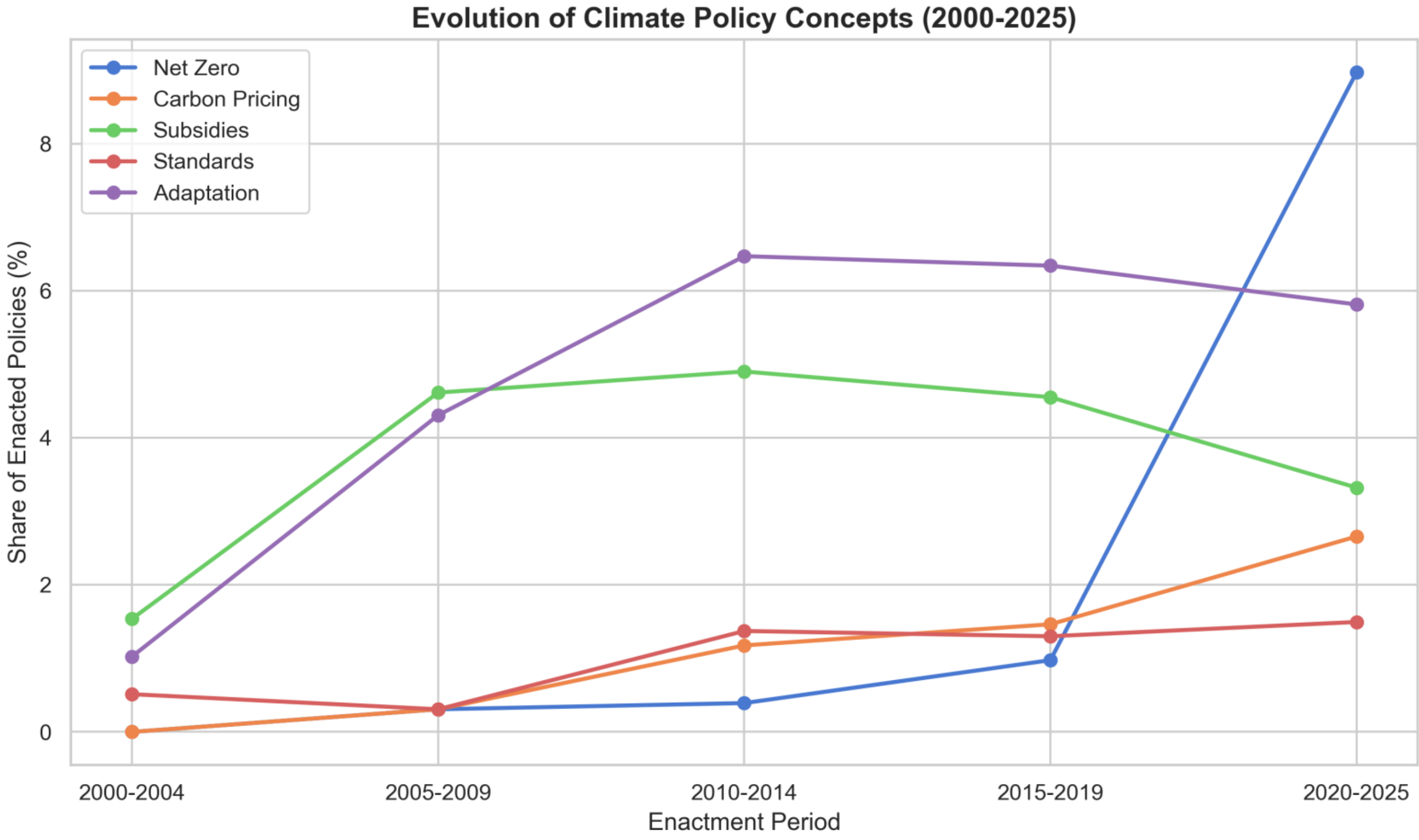
Appendix: Policy sequencing for one country-sector



The heatmap shows the likelihood of transitioning from one policy instrument to another. We’re trying to see if there is a certain pathway Targets > Strategies and plans > Binding instruments.

The first expected step is validated, but there is lower probability to move from strategy and planning to binding instruments.

Appendix: Evolution of climate policy concepts (2000-2025)



We chose **five key concepts** that serve as indicators for distinct policy eras and strategies:

1. Net Zero / Decarbonization (net_zero)
To track the shift from the pre-2015 era (focused on general "emissions reductions") to the post-Paris Agreement era (focused on absolute, long-term carbon neutrality).

2. Carbon Pricing (carbon_pricing)
To capture the rise of market-based instruments and economic debates around placing a direct price on carbon emissions.

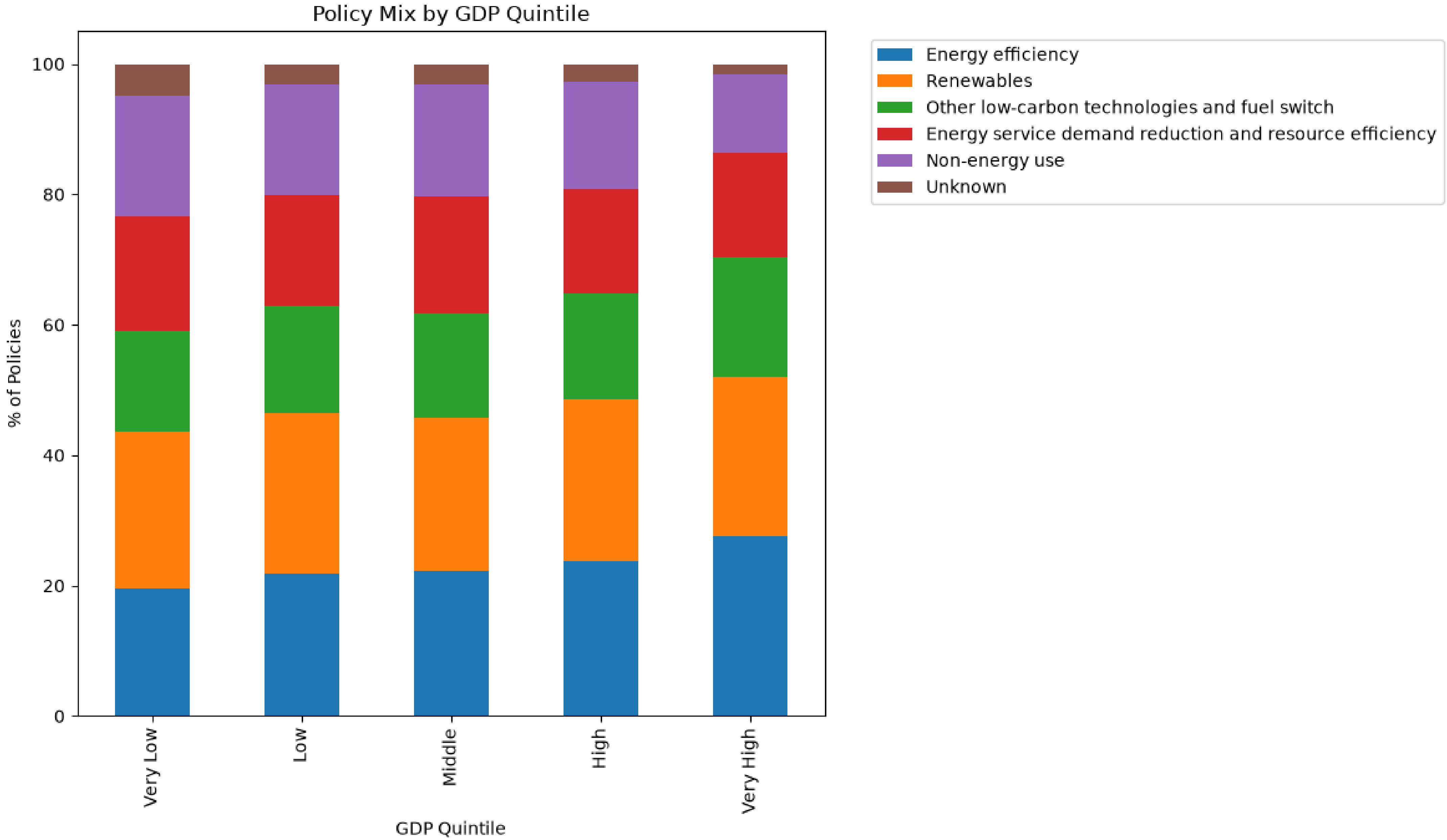
3. Subsidies & Incentives (subsidies)
To track fiscal support such as Feed-in Tariffs (FiTs) and green tax credits, which were the primary drivers used by governments to commercialize and scale up wind and solar technologies.

4. Standards & Mandates (standards)
To track "sticks" (command-and-control regulations) such as building codes and vehicle performance mandates, which force behavioral and engineering changes regardless of market prices.

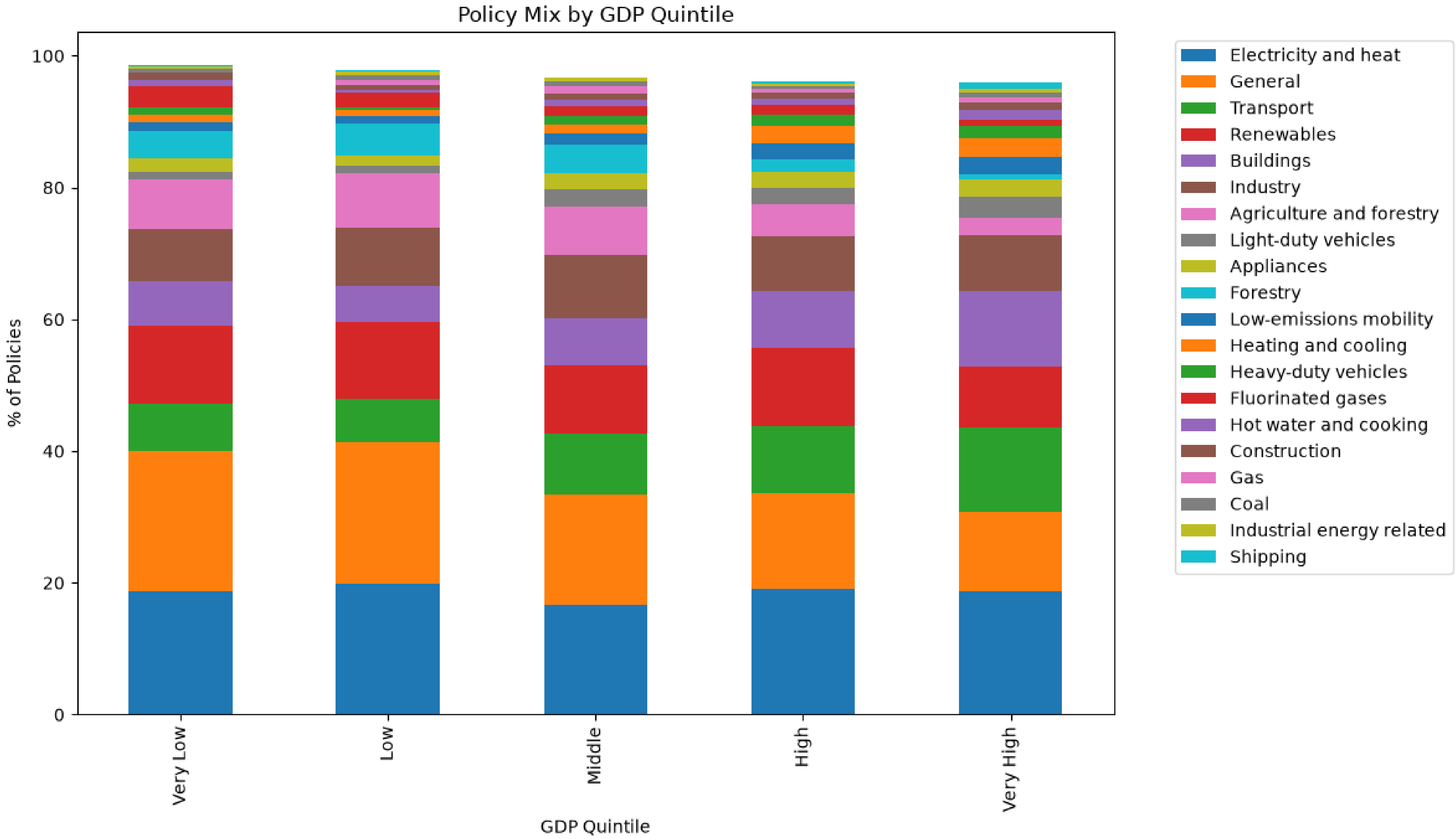
5. Adaptation & Resilience (adaptation)
To measure the growing recognition that mitigation alone is insufficient. This tracks the rise of policies addressing the physical impacts of climate change (disaster risk, vulnerability, and resilience), which is particularly vital for developing economies.

Findings: We observe a structural shift in policy terminology. Terms like "net zero" and "carbon pricing" have grown exponentially, with net zero appearing in **9.0%** of descriptions from 2020 to 2025. Trigrams show that Cluster 0 and 1 are dominated by international treaty compliance, while Cluster 2 focuses on Specific Phase timelines.

Appendix: Policy analysis by GDP (all countries) - energy



Appendix: Policy analysis by GDP (all countries) – policy type



Appendix: Policy analysis by GDP (all countries) – policy type vs instrument